
# ADDITIONAL PRODUCT INFORMATION
# Sikaflex®-265

13.11.2013 / VERSION 3 / SIKA SERVICES AG / MARKETFIELD ENGINEERING

**CHEMICAL AND MECHANICAL PROPERTIES ACCORDING E DIN 6701-3: 2010-8**

BUILDING TRUST

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# 1 CHEMICAL PROPERTIES

## 1.1 CHEMICAL BASE

| C1 | Chemical base | |
|---|---|---|
| | Polyurethane | See product data sheet |

## 1.2 CURING MECHANISM

| C2 | Curing mechanism | |
|---|---|---|
| | 1C Moisture curing | See product data sheet |

## 1.3 VOLATILE-ORGANIC COMPOUNDS

| C3 | Volatile Organic Compounds | |
|---|---|---|
| | < 0.01% VOC | See safety data sheet / Chapter 15 |

## 1.4 TOXICOLOGICAL PROPERTIES

| C4 | Toxicological properties | |
|---|---|---|
| | Classification (67/548/EEC) | See safety data sheet / Chapter 2.1 |

## 1.5 PH- VALUE

| C5 | pH Value | |
|---|---|---|
| | According E DIN 6701-3 | 6.5 |

## 2 MECHANICAL PROPERTIES

### 2.1 DENSITY UNCURED

| M1 | Density uncured |
|---|---|
| | See product data sheet |

### 2.2 COLOUR

| M3 | Colour |
|---|---|
| | black |

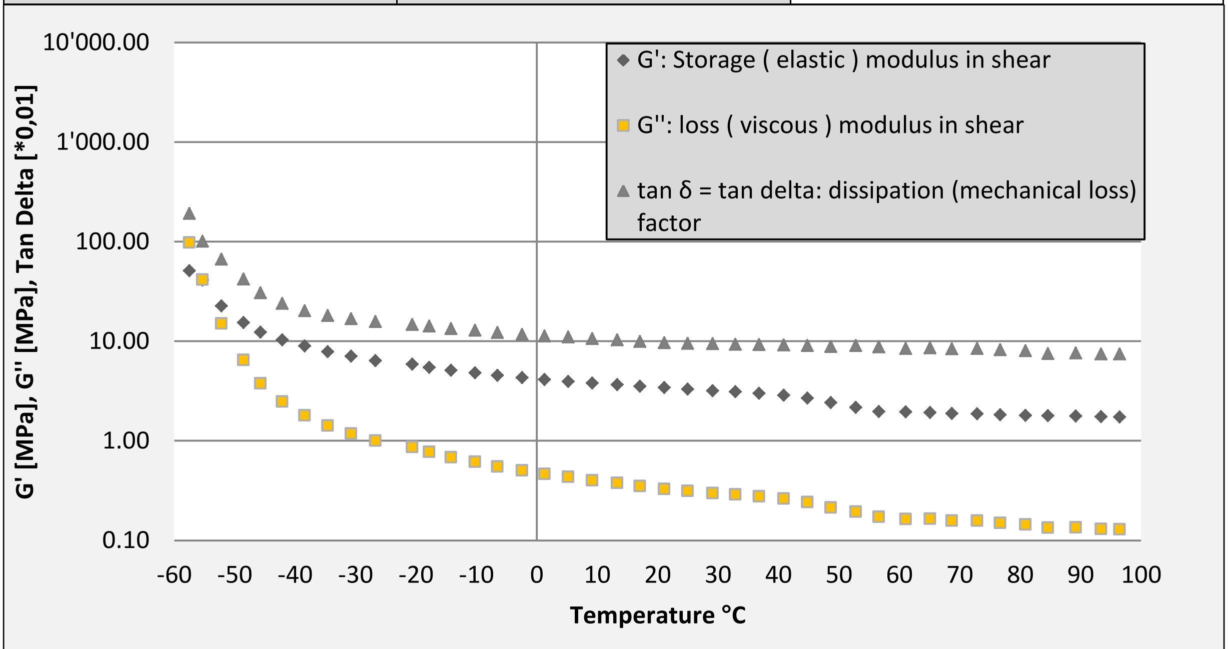
### 2.3 SHRINKAGE

| M5 | Shrinkage |
|---|---|
| | See product data sheet |

### 2.4 THERMO MECHANICAL BEHAVIOR & GLASS TRANSITION TEMPERATURE

| M7 | Thermo mechanical behavior & Glass transition temperature | E DIN 6701:2010-8 |
|---|---|---|

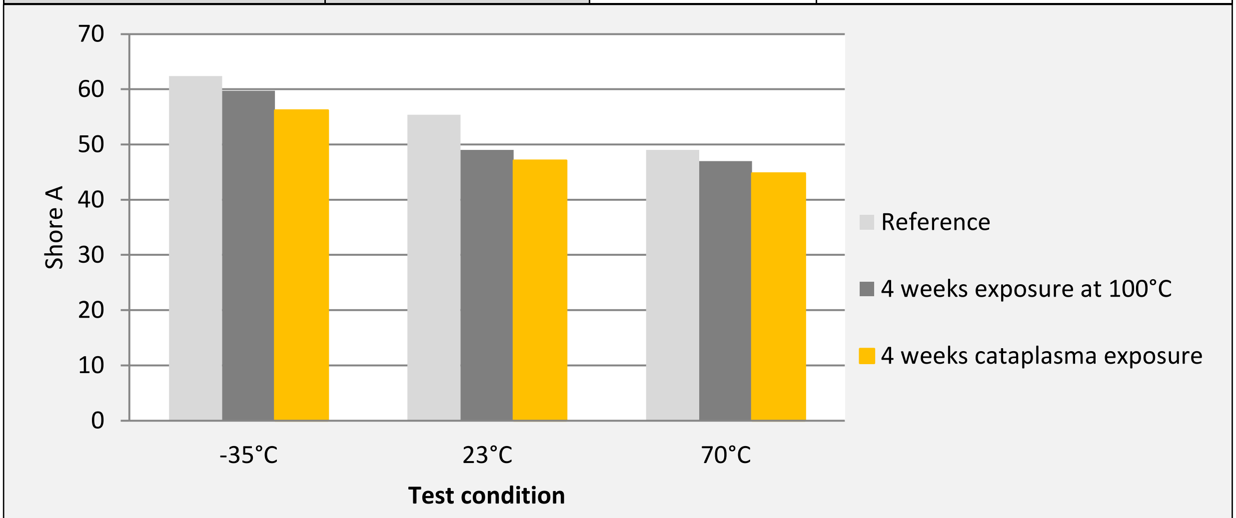
| Characteristics | Standard / Norm | Value / Tg |
|---|---|---|
| Glass transition temperature | DIN EN ISO 6721-3<br>Sika CQP 509-1 | approx. - 58°C |



## 2.5 SHORE A HARDNESS

| M8 | Shore A hardness | E DIN 6701:2010-8 |
|---|---|---|

| After curing, 7 days at 23°C/50%r.h. | | | |
|---|---|---|---|
| **Characteristics** | **Standard / Norm** | **Test condition** | **Value** |
| Shore A | DIN 53505<br>Sika CQP 023-1 | -35°C | 62 |
| | | 23°C | 55 |
| | | 70°C | 49 |
| After 4 weeks cataplasma ageing (70°C/100% r.h.) | | | |
| **Characteristics** | **Standard / Norm** | **Test condition** | **Value** |
| Shore A | DIN 53505<br>Sika CQP 023-1 | -35°C | 56 |
| | | 23°C | 47 |
| | | 70°C | 44 |
| After 4 weeks at 100°C | | | |
| **Characteristics** | **Standard / Norm** | **Test condition** | **Value** |
| Shore A | DIN 53505<br>Sika CQP 023-1 | -35°C | 59 |
| | | 23°C | 49 |
| | | 70°C | 47 |



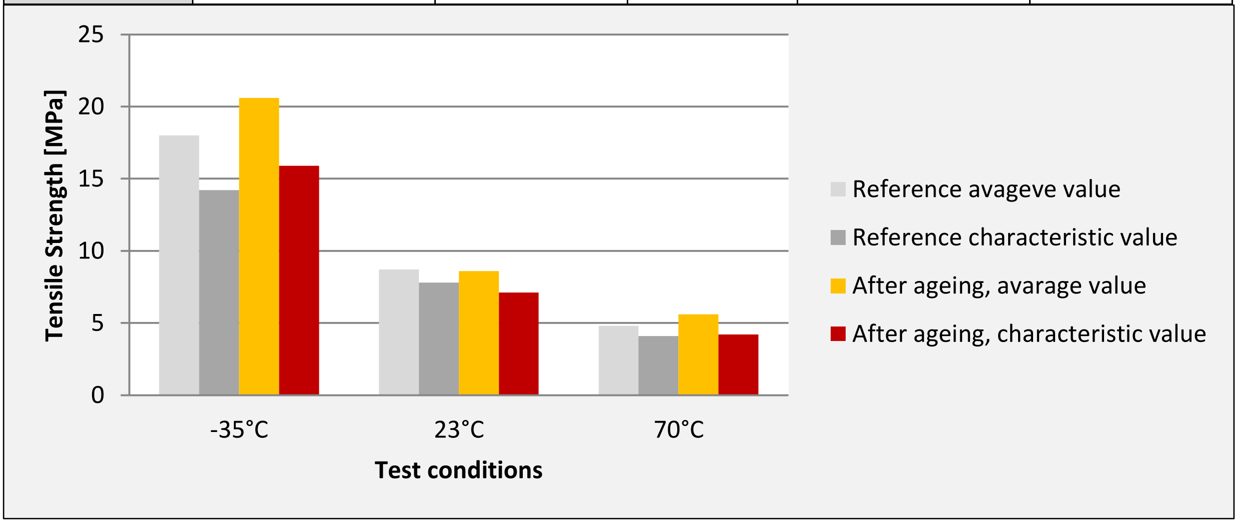
## 2.6 TENSILE STRENGTH / ELASTICITY / E-MODULUS

| M9 | Tensile Strength / Elasticity / E-Modulus | Quasi Static Test<br>E DIN 6701:2010-8 |
|---|---|---|

| Conditions | Standard average values | | | Characteristic value $R_C$ | |
|---|---|---|---|---|---|
| | Tensile Strength [$\sigma_{max}$] | Elasticity [$\%_{max}$] | E-Modulus [E] | Tensile Strength [$\sigma_{max}$] | Elasticity [$\%_{max}$] |
| -35°C | 18.3 MPa | 404% | 10.5 MPa | 14.2 MPa | 294% |
| 23°C / 50%r.h. | 8.7 MPa | 488% | 3.4 MPa | 7.8 MPa | 431% |
| 70°C | 4.8 MPa | 248% | 2.6 MPa | 4.1 MPa | 201% |

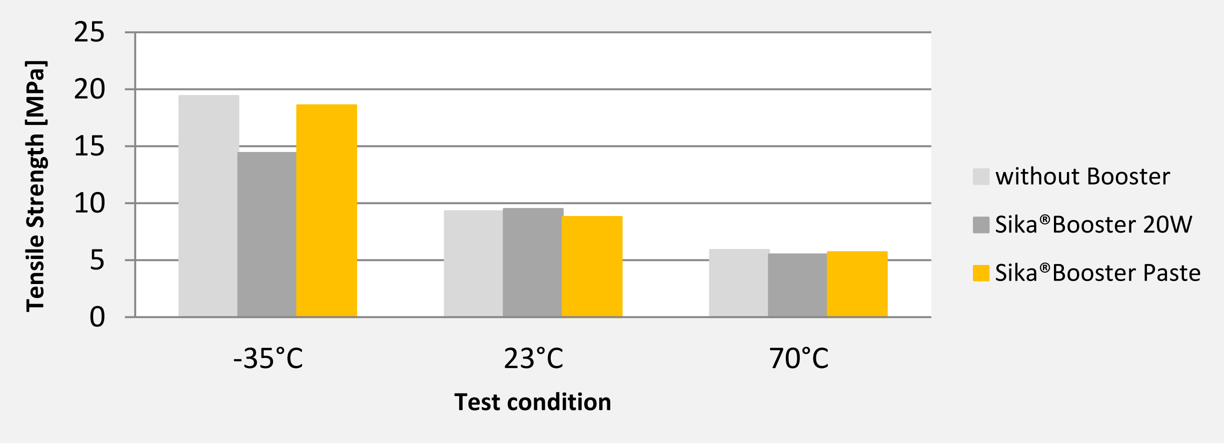
After ageing cycle according to DIN 54457
7d 23°C/50%r.h. + 7d water immersion + 1d at 80°C + 7d cataplasma (70°C/100%r.h.)

| Conditions | Standard average values | | | Characteristic value $R_C$ | |
|---|---|---|---|---|---|
| | Tensile Strength [$\sigma_{max}$] | Elasticity [$\%_{max}$] | E-Modulus [E] | Tensile Strength [$\sigma_{max}$] | Elasticity [$\%_{max}$] |
| -35°C | 20.6 MPa | 377% | 18.2 MPa | 15.9 MPa | 268% |
| 23°C / 50%r.h. | 8.6 MPa | 403% | 2.6 MPa | 7.1 MPa | 358% |
| 70°C | 5.6 MPa | 266% | 2.2 MPa | 4.2 MPa | 180% |



## 2.7 COMPARISION OF MECHANICAL PROPERTIES WITH AND WITHOUT BOOSTER PASTE

| | | Standard average values | | |
|---|---|---|---|---|
| **System** | **Test condition** | **Tensile Strength [σ max]** | **Elasticity [% max]** | **E-Modulus [E]** |
| Reference (without Booster) | - 35°C | 19.4 MPa | 430% | 7.6 MPa |
| Sika®Booster 20W | | 14.5 MPa | 337% | 8.3 MPa |
| Sika®Booster Paste | | 18.6 MPa | 455% | 7.2 MPa |
| Reference (without Booster) | 23°C | 9.3 MPa | 487% | 3.9 MPa |
| Sika®Booster 20W | | 9.5 MPa | 496% | 3.9 MPa |
| Sika®Booster Paste | | 8.8 MPa | 447% | 3.7 MPa |
| Reference (without Booster) | 70°C | 5.9 MPa | 292% | 3.2 MPa |
| Sika®Booster 20W | | 5.5 MPa | 286% | 3.0 MPa |
| Sika®Booster Paste | | 5.7 MPa | 277% | 3.2 MPa |

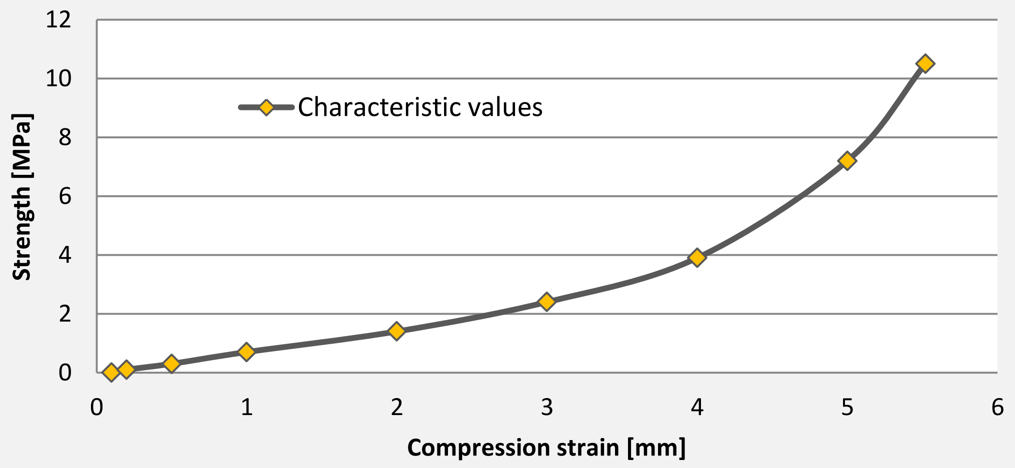



Based on the results listed above we can confirm that the influence of the booster paste on mechanical properties is negligible.

## 2.8 COMPRESSION STRENGTH

| M10 | Compression Strength | Quasi Static Test<br>E DIN 6701:2010-8 |
|---|---|---|

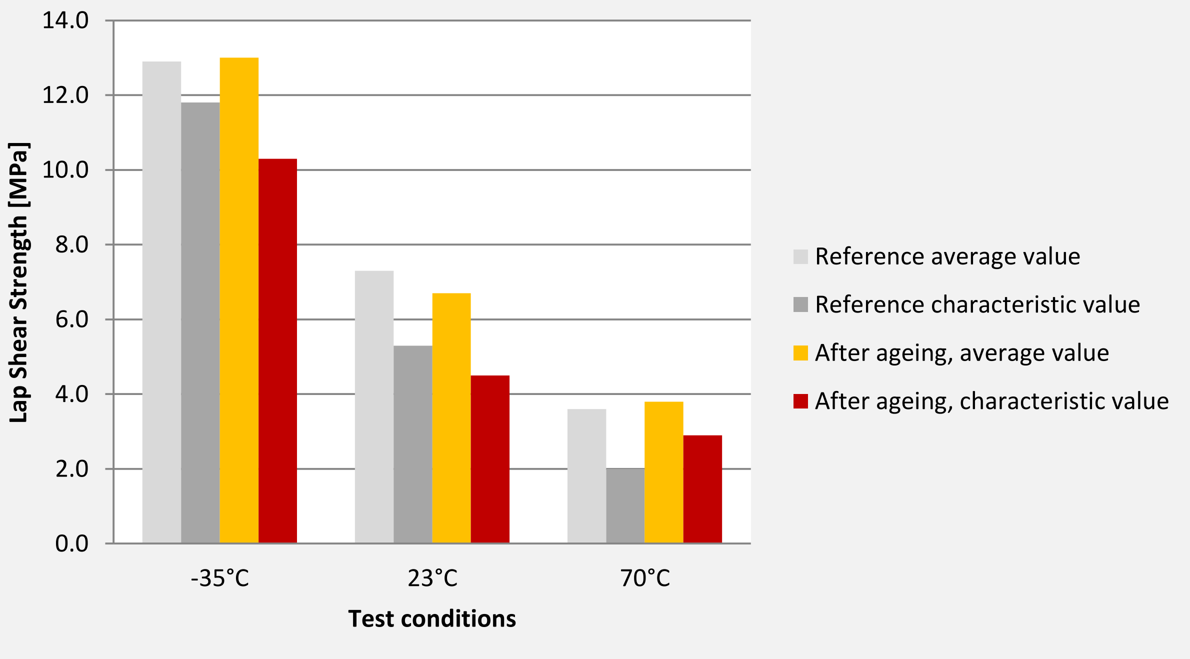
| Test Condition | Deformation [mm] | Compression stress [σ] | Compression force [N] |
|---|---|---|---|
| 23°C | 1 | 0.7 Mpa | 201 N |
| | 2 | 1.4 Mpa | 408 N |
| | 3 | 2.4 Mpa | 682 N |
| | 4 | 3.9 Mpa | 1135 N |
| | 5 | 7.2 Mpa | 2108 N |
| | 5.5 | 10.5 Mpa | 3147 N |
| Sample dimension (cylinder): 10mm height, 20mm diameter | | | |



## 2.9 LAP SHEAR STRENGTH

| M11 | Lap Shear Strength | Quasi Static Test<br>E DIN 6701:2010-8 |
|---|---|---|

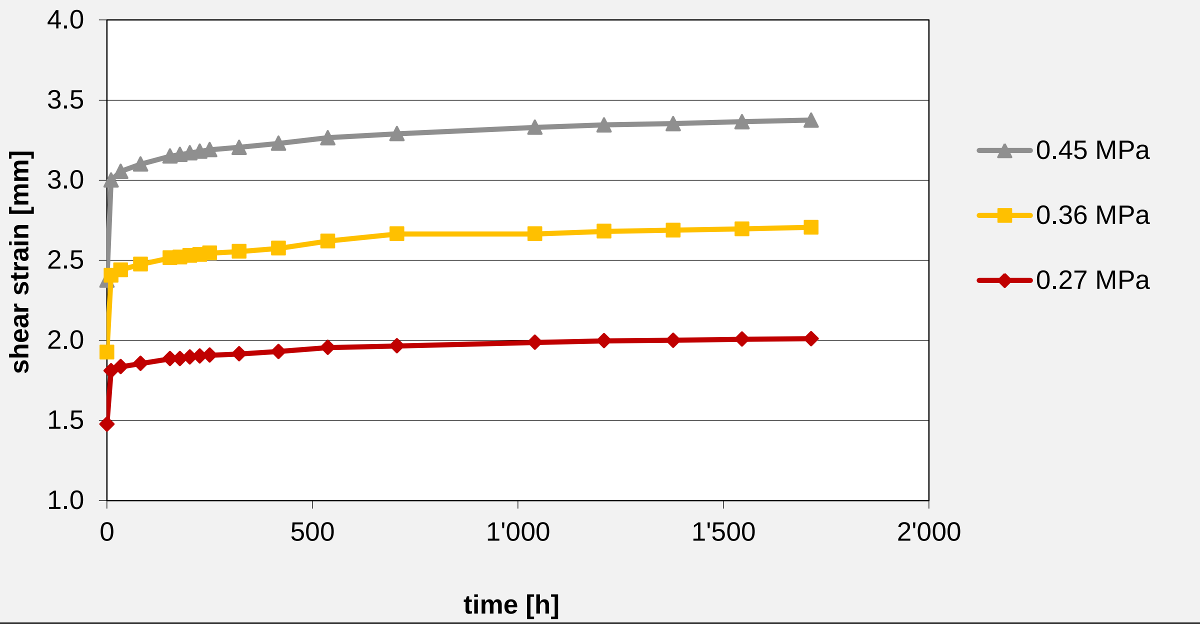
| | Standard average values | | Characteristic value $R_C$ | |
|---|---|---|---|---|
| Conditions | Lap shear strength [$\tau_{max}$] | Shear strain at break [$\gamma_{max}$] | Lap shear strength [$\tau_{max}$] | Shear strain at break [$\gamma_{max}$] |
| -35°C | 12.9 Mpa | 425% | 11.8 Mpa | 256% |
| 23°C / 50%r.h. | 7.2 Mpa | 496% | 5.3 Mpa | 342% |
| 70°C | 3.6 Mpa | 338% | 2.0 Mpa | 142% |
| After ageing cycle according to DIN 54457<br>7d 23°C/50%r.h. + 7d water immersion + 1d at 80°C + 7d cataplasma (70°C/100%r.h.) | | | | |
| | Standard average values | | Characteristic value $R_C$ | |
| Conditions | Lap shear strength [$\tau_{max}$] | Shear strain at break [$\gamma_{max}$] | Lap shear strength [$\tau_{max}$] | Shear strain at break [$\gamma_{max}$] |
| -35°C | 13.0 Mpa | 431% | 10.3 Mpa | 239% |
| 23°C / 50%r.h. | 6.7 Mpa | 425% | 4.5 Mpa | 164% |
| 70°C | 3.8 Mpa | 258% | 2.9 Mpa | 143% |
| Dimension of bond line: overlap = 12mm, width = 25mm, thickness = 3mm | | | | |



## 2.10 CREEP

| M12 | Creep | Static Test<br>E DIN 6701:2010-8 |
|---|---|---|

| Test condition | Static load | Deformation [mm] | Deformation [%] |
|---|---|---|---|
| 90 days at 23°C | 6.75 Kg / 0.27 MPa | 2.0 mm | 40% |
| | 13.5 Kg / 0.36 MPa | 2.7 mm | 54% |
| | 22.5 Kg / 0.45 MPa | 3.4 mm | 68% |
| Dimension of bond line: overlap = 20mm, width = 25mm, thickness = 5mm | | | |




* 0.27 MPa ≅ 6% of lap shear strength value stated in the product data sheet of Sikaflex®-265

## 2.11 RELAXATION SHEAR FAILURE STRAIN

| M13 | Relaxation Shear Failure Strain | Static Test<br>E DIN 6701:2010-8 |
|---|---|---|

| Characteristics | Test condition | Deformation | Assessment |
|---|---|---|---|
| Allowable shear strain | 1000 hours at 23°C | (tan $\gamma$) 0.5<br>**50% = 2.5mm** | **Visual assessment OK**<br>**NO FAILURE** |

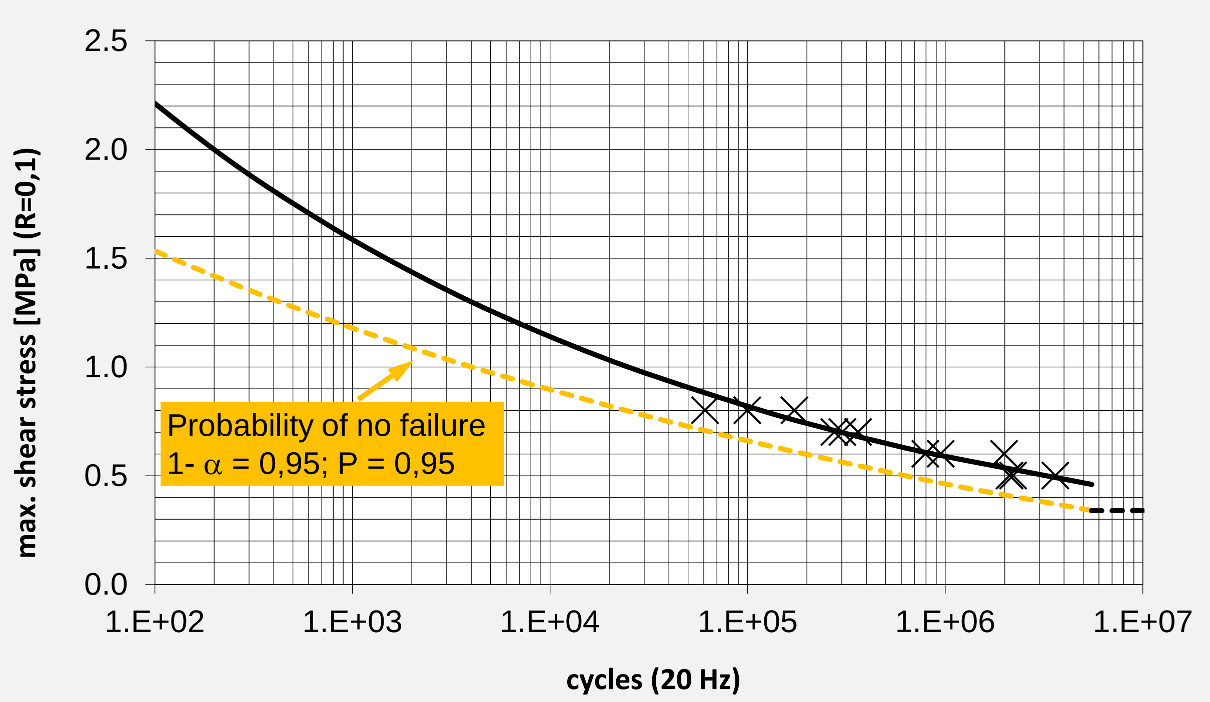
| After ageing cycle according to DIN 54457<br>7d 23°C/50%r.h. + 7d water immersion + 1d at 80°C + 7d cataplasma (70°C/100%r.h.) | | | |
|---|---|---|---|
| **Characteristics** | **Test condition** | **Deformation** | **Assessment** |
| Allowable shear strain | 1000 hours at 23°C<br>+<br>ageing cycle | (tan $\gamma$) 0.5<br>**50% = 2.5mm** | **Visual assessment OK**<br>**NO FAILURE** |
| Dimension of bond line: overlap = 20mm, width = 25mm, thickness = 5mm | | | |

## 2.12 SHEAR FATIGUE TESTING

| M14 | Shear Fatigue Testing | Dynamic Test |
|---|---|---|

| Characteristics | Test condition | Cycles | Nominal shear stress amplitude at cycles to failure |
|---|---|---|---|
| 95% Statistical probability of no failure | 23°C | $10^7$ | 0.34 Mpa |

Frequency: According to E DIN 6701:2010-08 Appendix A / Section 1.3.7
Amplitude ratio value / R = 0.1
stress regulated amplitude



# 3 REFERENCES

| Pos. | Report no: / Source | Title |
|---|---|---|
| C1 | Product data sheet | Chemical base |
| C2 | Product data sheet | Curing mechanism |
| C3 | Safety data sheet | VOC content |
| C4 | Safety data sheet | Toxicological properties |
| C5 | 00506-CTS1-00005-SKh | pH Values of Sikaflex® adhesives according to E DIN 6701-3 |
| | | |
| M1 | Product data sheet | Density uncured |
| M3 | Product data sheet | Colour |
| M5 | Product data sheet | Shrinkage |
| M7 | 00509-CTS1-00017-MKö | Thermo mechanical behaviour & glass transition temperature |
| M8 | 00509-CTS1-00004-MKö | Shore A hardness of Sikaflex®-265 according to E DIN 6701-3 |
| M9 | 00509-CTS1-00011-MKö | Tensile strength of Sikaflex®-265 acc. To E DIN 6701-3 |
| | 00509-CTS1-00044-MKö | Influence of Booster Paste on mechanical properties of Sikaflex®-265 |
| M10 | 00509-CTS1-00043-MKö | Compressive strength Sikaflex®-265 acc. to E DIN 6701-3 |
| M11 | 00509-CTS1-00023-MKö | Lap shear strength of Sikaflex®-265 acc. to E DIN 6701-3 |
| M12 | 00509-CTS1-00029-BVs | Creep behavior of Sikaflex®-265 acc. to E DIN 6701-3 |
| M13 | 00509-CTS1-00036-BVs | Relaxation test of Sikaflex®-265 following E DIN 6701-3 |
| M14 | 2013-05-01-MM-SKo | Fatigue Test of Sikaflex®-265 |



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